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The effect of external nutrient resources on the population dynamics of zooxanthellae in a reef coral

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Experiments were done to determine if ammonium, phosphate and feeding on *Artemia* nauplii affected the population density of symbiotic algae (zooxanthellae) in the Red Sea coral *Stylophora pistillata*. Corals were incubated for 14 days under natural sunlight at reduced intensity in running seawater aquaria. The seawater was continuously spiked to give final concentrations of either 20 μM ammonium or 2 μM phosphate, or both. A second set of similarly treated corals was also fed *Artemia* nauplii daily. Population density of zooxanthellae in corals spiked with ammonium, or ammonium plus phosphate, approximately doubled, and the ratio of zooxanthellae carbon:nitrogen decreased. Phosphate supplementation alone had no effect. The increase in zooxanthellae numbers was linearly proportional to the increase in protein in zooxanthellae, suggesting that availability of inorganic nitrogen leads to increased protein synthesis in zooxanthellae. Feeding on *Artemia* alone or together with phosphate had no effect on the population density of zooxanthellae. Feeding on *Artemia* and ammonium produced a small increase in population density but it was not statistically significant. The small effect could be due to insufficient influx of ammonium in fed animals, or growth of both animal and algae resulting in little or no net change in the population density of zooxanthellae. The results are consistent with the hypothesis that the growth of zooxanthellae in *S. pistillata* from the Red Sea is nitrogen limited.

Introduction

Symbiotic dinoflagellates ('zooxanthellae') contribute to the growth and maintenance of many reef corals and other tropical cnidarians (Muscatine 1980*a*). They are major primary producers that translocate reduced organic carbon to their host. They also take up dissolved inorganic nutrients, which, together with nutrients obtained from food, are retained and recycled within the association (Muscatine & Porter 1977; Falkowski *et al.* 1984). Since the initial work of

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Drew (1972), it has been observed repeatedly that the density per unit area of zooxanthellae in coral tissue remains relatively constant, at approximately 10^6 to 2×10^6 cells per square centimetre (Muscatine *et al.* 1985), suggesting that the population density of zooxanthellae might be regulated. It was therefore of interest to determine if environmental factors, such as light and nutrients, have any effect on density per unit area of zooxanthellae.

In previous studies (Dubinsky *et al.* 1984; Porter *et al.* 1984; Muscatine *et al.* 1984) we investigated the influence of irradiance on growth and primary production of zooxanthellae in the Red Sea coral *Stylophora pistillata*. We noted that the population density of zooxanthellae in shallow-water *S. pistillata* was about 1.5×10^6 cells per square centimetre, and that about 90% of the carbon fixed photosynthetically by zooxanthellae was translocated to the host, regardless of growth irradiance. The zooxanthellae adapted to decreased irradiance by altering the amount of chlorophyll *a* per cell (Falkowski *et al.* 1984), and it was only at attenuated irradiance at depth that we observed a decrease in the cell density of zooxanthellae (McCloskey & Muscatine 1984). From these observations we concluded that populations of zooxanthellae in shallow-water *S. pistillata* are 'light sufficient', and able to supply the carbon required for respiration and growth of the coral. However, translocated carbon is nitrogen deficient and is largely respired (Battey & Patton 1987), and only about 8% of the total carbon fixed and translocated in *S. pistillata* is assimilated by the animal (Falkowski *et al.* 1984). These observations focused attention on the possibility that growth of zooxanthellae in *S. pistillata* might be nutrient limited.

There have been many studies demonstrating the uptake and assimilation of dissolved inorganic nitrogen (DIN) (Muscatine 1980*b*; Summons & Osmond 1981; Burris 1983, 1984; Wilkerson & Trench 1986; Summons *et al.* 1986; Anderson & Burris 1987) and phosphorus (D'Elia 1977) by zooxanthellae in symbiotic cnidarians, including corals. The effect of DIN on zooxanthellae productivity was investigated by Taylor (1978) who reported that high concentrations (20–200 μM) of ammonium and urea caused an increase in the photosynthetic rate of zooxanthellae in the coral *Acropora cervicornis*. The phenomenon was confirmed by Summons *et al.* (1986), who reported that 25 μM ammonium increased the rate of photosynthesis in some populations of *Stylophora pistillata* from the Great Barrier Reef.

The effect of various dissolved and particulate nutrients on zooxanthellae population density has also been investigated. Meyer & Schultz (1985*a*, *b*) described the localized defecation and excretion by haemulid fishes over colonies of *Porites furcata* and *Acropora palmata*. Excretory products were rich in nitrogen (as ammonium) and phosphorus, and concomitant increases in coral tissue nitrogen and phosphorus, tissue biomass, and density of zooxanthellae, may have resulted from localized nutrient enrichment. Smith & Muscatine (1986) found that, over a test period of 2 weeks, pulsed addition of ammonium and phosphate did not affect the diel pattern of mitotic index (MI) of zooxanthellae in starved polyps of the sea anemone *Aiptasia pulchella*, but feeding had a pronounced stimulatory effect. Feeding produced similar results in starved *A. pallida* from Bermuda but addition of inorganic N and P caused a transient increase in MI of zooxanthellae (Cook *et al.* 1988; see also Cook & D'Elia 1987).

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We report here the results of controlled experiments to determine if ammonium and phosphate, as well as feeding on nauplii of the brine shrimp *Artemia* sp., affect the population density of zooxanthellae in the Red Sea coral *Stylophora pistillata*. Companion studies were done to gain insight into dynamics of nitrogen recycling (Rahav *et al.* 1989), and the effects of nutrient enrichment on primary production and respiration in this coral species.

Methods

Collection and maintenance of corals

All experiments were done at the Interuniversity Marine Laboratory, Eilat, Israel, in February 1986 and March 1987. Colonies of *Stylophora pistillata* (light-adapted (Falkowski & Dubinsky 1981)) of roughly similar size were collected from 2–5 m on the reef adjacent to the marine laboratory. They were briefly maintained in running seawater while unwanted material was trimmed above the base of each colony and ectocommensal crabs and shrimps were removed.

Incubation with nutrients and Artemia

An aquarium constructed from clear Plexiglas was partitioned into 10 individual chambers (28.5 cm × 19.3 cm × 20 cm). Each chamber was supplied with running sea water from the laboratory system at a flow rate of 30–36 l h^{-1}. Overflow drains in the aquarium maintained a volume of 8855 ml per chamber. The seawater system was fed by ambient seawater that percolates through a sand filter into a large reservoir. Although surface waters of the Gulf of Eilat often contain about 0.1 μM phosphate, the concentrations of ammonium, nitrate and phosphate in the ambient seawater used in these experiments were below the limits of detection by conventional chemical tests.

Eight of the chambers were utilized as follows. Six black plastic reservoirs were filled with seawater and duplicate pairs were spiked with either ammonium (as ammonium sulphate), phosphate (as sodium dihydrogen phosphate), or ammonium plus phosphate to yield final concentrations of 2 mM ammonium and 200 μM phosphate. These spiked solutions were siphoned from the reservoirs into the aquarium chambers at a constant head pressure and a flow rate of 5.8 ml min^{-1}. Final chamber concentrations were 20 μM ammonium (N), 2 μM phosphate (P), or N + P in these same concentrations. Two chambers received no nutrient supplement. One set of four aquarium chambers was supplied once daily with *Artemia* nauplii (Art). The final eight chambers were designated as control (untreated), N, N + P, P, *Artemia*, *Artemia* + N + P, *Artemia* + N, and *Artemia* + P. Nutrient flow rates were checked periodically, and all chambers were cleaned as necessary to minimize the growth of ambient algae. Three colonies of *S. pistillata* were placed in each chamber. The chamber aquarium was exposed to natural sunlight but was screened to adjust maximum irradiance to approximately 1000 μmol quanta m^{-2} s^{-1}. Incubations ran continuously for 14 days.

Analytical procedures

After incubation for 14 days tissue was removed from all corals with a Water-Pik (Johannes & Wiebe 1970). The surface area of the skeleton was measured by

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subdividing it into small pieces, measuring the projected area of the pieces on a leaf-area image analyser (Delta Devices), and assuming the pieces to have a sub-cylindrical shape (Falkowski & Dubinsky 1981).

The volume of the resulting tissue homogenate was recorded and samples were taken for determination of zooxanthellae numbers (haemocytometer), mitotic index and specific growth rate (Wilkerson *et al.* 1983), and chlorophyll *a*, c_2 (Jeffrey & Humphrey 1975). Homogenates were also analysed for total organic carbon and nitrogen (C, N) (Perkin–Elmer 240B Elemental Analyzer), and total carbohydrate, lipid and protein. Biochemical determinations were based on the micro-analytical technique described by Holland & Gabbott (1971), except that carbohydrates were assayed by the phenol-sulphuric acid method of Strickland & Parsons (1972) and lipids were assayed by the method of Marsh & Winstein (1966). The homogenate was centrifuged to separate algae and animal tissue, and each fraction was analysed for C, N, total protein, carbohydrate and lipid.

Results

The effects of incubation for 14 days with inorganic nutrients and *Artemia* on coral biomass parameters (per square centimetre) are presented in table 1.

Zooxanthellae densities ranged from 1.11×10^6 for the untreated corals to 2.22×10^6 for N+P-treated corals. One-way ANOVA, after logarithmic transformation of the data, shows differences in zooxanthellae density among treatments at the 90% confidence level. Further analysis, using the least significant difference (LSD) multiple comparison method (Sokal & Rohlf 1981), reveals that zooxanthellae densities of only the corals treated with N and N+P were significantly higher than the untreated controls. Densities of zooxanthellae in corals treated with N+P were also significantly higher than those in corals treated with P and *Artemia*. These results suggest that addition of ammonium caused a significant increase in zooxanthellae density.

Chlorophyll *a* in N- and N+P-treated corals (6.93–7.34 μg cm^{-2}) differed significantly from the untreated corals (2.61 μg cm^{-2}) at the 95% confidence level. Analysis by LSD demonstrated also that chlorophyll *a* content per square centimetre in N+P-treated corals was similar only to that of N-treated corals, and was significantly higher than that in untreated, P-, *Artemia*-, and *Artemia*+N+P-treated corals.

Table 1 also indicates that addition of DIN caused a decrease in C:N ratios of coral tissue from 7.3 to as low as 5.67, but this decrease is not significantly different from changes caused by other treatments ($p > 0.90$). Analysis of tissue carbohydrate, lipid and protein revealed no trends correlative with treatments.

The results of the various treatments on zooxanthellae parameters are presented in table 2. Chlorophyll *a* (picograms per cell) ranges from 2.19 for *Artemia*+N+P- to 3.58 for N-treated corals. Although the increase in average chlorophyll *a* per cell is highest in N- and N+P-treated corals, none of the treatments caused changes significantly different from controls ($p > 0.90$). Thus the increase in chlorophyll *a* per square centimetre (table 1) was due primarily to increased numbers of zooxanthellae and not simply to an increase in chlorophyll *a* per cell.

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Table 1. Whole coral parameters (per square centimetre) as means ± standard error of means

treatment		10^{-6} × cells	chl *a*/μg	C/μg	N/μg	C:N (mol/mol)	CHO/mg	lipid/mg	total protein/mg	animal protein/mg
untreated	mean	1.11	2.61	704.7	112.7	7.31	0.20	0.83	1.49	1.34
	s.e.m.	0.56	1.01	52.8	5.9	0.16	0.09	0.32	0.36	0.39
		$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
N	mean	1.96	6.93	766.3	152.2	5.88	0.17	0.75	2.30	1.98
	s.e.m.	0.31	1.15	118.2	3.4	0.78	0.00	0.08	0.36	—
		$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 1$
N+P	mean	2.22	7.34	844.2	172.4	5.80	0.24	0.88	2.16	1.80
	s.e.m.	0.44	1.32	60.2	21.4	1.13	0.04	0.19	0.30	0.32
		$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
P	mean	1.28	3.03	493.1	86.2	6.69	0.18	0.58	1.62	1.38
	s.e.m.	0.36	0.42	141.2	0.8	1.86	0.04	0.01	0.45	0.43
		$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
Art	mean	1.24	2.96	371.1	59.5	6.84	0.17	0.66	1.46	1.29
	s.e.m.	0.27	1.12	32.4	—	—	0.03	0.15	0.61	0.56
		$n = 3$	$n = 3$	$n = 2$	$n = 1$	$n = 1$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
Art+N+P	mean	1.68	3.71	518.3	100.2	5.57	0.20	0.79	2.09	1.77
	s.e.m.	0.11	0.42	0.59	—	—	0.03	0.05	0.23	0.21
		$n = 2$	$n = 2$	$n = 2$	$n = 1$	$n = 1$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
Art+N	mean	1.55	3.96	524.4	95.7	6.4	0.21	0.68	2.28	1.84
	s.e.m.	0.12	0.94	69.5	9.8	0.2	0.05	0.17	0.43	0.52
		$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
Art+P	mean	1.43	4.12	640.2	106.2	7.12	0.20	0.81	2.42	2.13
	s.e.m.	0.31	2.23	187.4	36.1	0.36	0.02	0.08	0.98	0.97
		$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$

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Table 2. Zooxanthellae parameters (per cell) as means ± standard errors of means

treatment	chl *a* / pg	chl *a*:chl *c* (mol/mol)	C/pg	N/pg	C:N (mol/mol)	C:chl *a* (by mass)	CHO/pg	lipid/pg	protein/pg	MI (%)	*u* (per day)
untreated	2.43	2.88	183.8	26.9	7.7	90.4	33.7	79.6	116.8	3.52	0.08
	0.50	0.22	42.2	5.8	0.7	17.2	11.7	31.8	26.2	2.03	0.04
	$n = 3$	$n = 3$	$n = 4$	$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 3$	$n = 3$	
N	3.58	3.24	146	38.5	4.4	42.7	6.7	76.4	250	3.02	0.06
	0.53	0.3	12.6	3	0.2	9.1	1.2	4.4	—	0.85	0.02
	$n = 3$	$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 1$	$n = 3$	$n = 3$
N+P	3.32	3.1	136.6	33.6	4.7	42.4	19.3	65.1	163	3.77	0.08
	0.07	0.29	7.1	0.4	0.2	1.9	6.9	2.6	32.6	1.69	0.04
	$n = 3$	$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 3$	$n = 3$
P	2.46	2.64	161.8	26.1	6.3	73.6	22	111.2	193.2	3.37	0.07
	0.64	0.28	31.2	—	—	10.2	15.9	73.5	59.5	2.76	0.06
	$n = 3$	$n = 3$	$n = 2$	$n = 1$	$n = 1$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 3$	$n = 3$
Art	2.74	3.4	143.4	25.3	6.7	55.6	26.6	46.1	135.2	2.43	0.05
	1.06	0.28	1.4	3.7	0.9	26.6	3.3	12.9	4.1	1.2	0.03
	$n = 3$	$n = 3$	$n = 3$	$n = 3$	$n = 3$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 3$	$n = 3$
Art+N+P	2.19	3.18	144.7	38.2	5.0	66.8	22.6	61.5	191.4	2.45	0.05
	0.41	0.43	9.4	8.9	0.1	8.2	1.4	4	27.5	1.54	0.03
	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
Art+N	2.54	3.56	181.1	44.8	4.7	73.8	30.2	70.2	291.8	1.2	0.03
	0.42	0.69	45.3	6.4	0.5	30.2	2.8	2.6	79.2	0.69	0.01
	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$
Art+P	2.78	3.08	181.4	35.2	5.9	74.3	27.1	94.2	208.2	1.96	0.04
	0.96	0.11	73.5	8.1	1.1	52.1	0.1	14.8	50.6	1.9	0.04
	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$	$n = 2$

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Similarly, chlorophyll $a:c_2$ ratios ranged from 2.64 to 3.56 ($\bar{x} = 3.14$) and did not differ significantly between treatments. Analysis of the relative abundance of carbohydrates, lipids and proteins in the zooxanthellae (table 1) reveals that carbohydrate per cell is significantly lower for N-treated corals ($p < 0.05$).

The mean MI of zooxanthellae was 2.72 % and there was no significant difference in MI among the treatments.

Multiple correlation analysis on the logarithmic transformed data revealed correlations between C:N ratios in zooxanthellae and cells per area ($r = -0.87$); N per cell and protein per cell ($r = 0.87$); cells per area and chlorophyll per area ($r = 0.88$), and percentage zooxanthellae lipid and percentage zooxanthellae protein ($r = -0.98$). Closer inspection of the relation between C:N and cells per area (figure 1) shows that N-, N+P-, *Artemia*+N-, and *Artemia*+N+P-treated corals generally cluster with low C:N ratios and higher cell densities. C:N ratios converge to a lower limit of 4.2 (moles per mole), which is lower than the Redfield ratio of 5.68 (moles per mole). These data suggest that an increase in the availability of nitrogen results in an increase in cell numbers and a decrease in C:N ratios of the zooxanthellae.

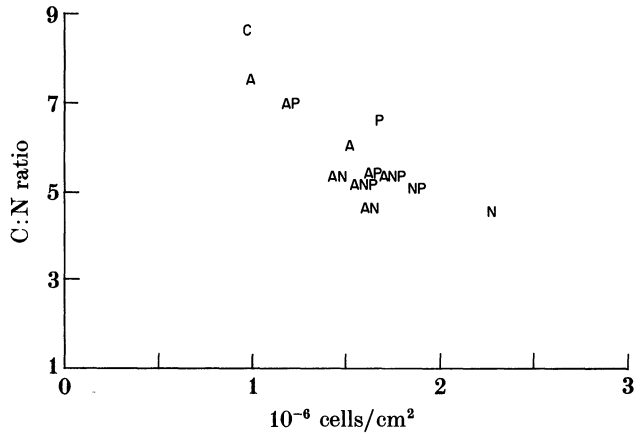

FIGURE 1. Zooxanthellae C:N ratios against density for each treatment, including control (C). Data in this and other figures are limited to paired data sets from the same coral colony. (Correlation coefficient $r = -0.78$.)

The relation between chlorophyll *a* and cells per unit area is shown in figure 2. The distribution of the data with respect to treatment again reveals that, in general, corals supplemented with N have greater cell densities and hence greater chlorophyll *a* concentrations. Further, the linear relation between chlorophyll *a* density and cell density as well as the poor correlation between chlorophyll *a* per cell and cells per square centimetre ($r = 0.35$) suggests that self-shading is not a significant factor contributing to the variation in chlorophyll *a* concentrations. These data provide further evidence that the availability of N strongly influences the concentration of cells and chlorophyll *a*.

The relation between N per cell and protein per cell is shown in figure 3. The increased cellular N is linearly proportional to cellular protein, with a slope of

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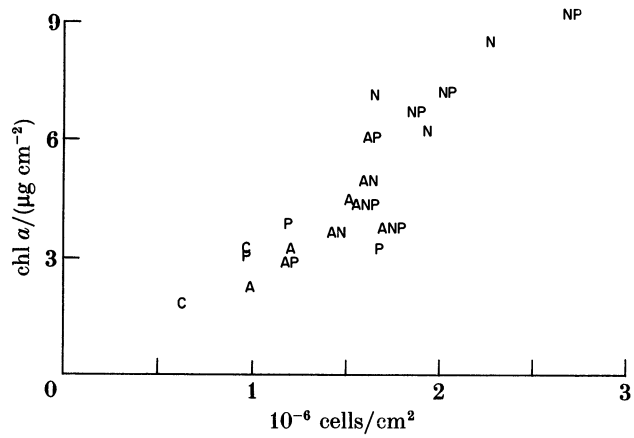

FIGURE 2. Chlorophyll per square centimetre against zooxanthellae density for each treatment, including controls ($r = 0.88$).

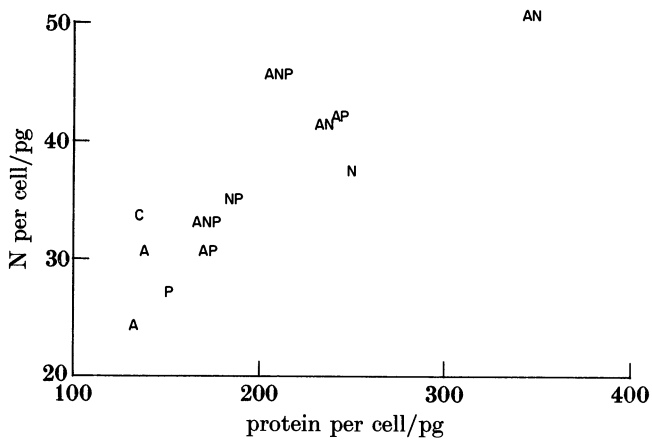

FIGURE 3. Nitrogen per cell against protein per cell for each treatment including controls ($r = 0.87$).

7.20 ± 1.25. This slope is not significantly different from the commonly accepted conversion factor of 6.25 μg N per microgram of protein. The correlation between N and protein suggests that an increased availability of N leads to an increased accumulation of zooxanthellae protein.

DISCUSSION

The results of this investigation show that exposure of *S. pistillata* for 14 days to 20 μM ammonium caused a significant increase in the population density of zooxanthellae. The increase is significantly greater than that produced by exposure to dissolved phosphate or particulate food (*Artemia*), either alone or in combination. These results are consistent with the set of predictions given in table 3. The predictions are based on the assumption that inorganic nitrogen is the nutrient

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Table 3. Predicted and observed effects of various resources on population density of zooxanthellae

resource	effect on growth of each component: algae	effect on growth of each component: animal	change in zooxanthellae population density: predicted	change in zooxanthellae population density: observed
N				
N+P	growth	no growth	increase unless expelled	increase
Art+N				
Art+N+P	growth	growth	none	none
Art				
Art+P	no growth	growth	none, if enough N from *Artemia* or else decrease	none

most often limiting the accumulation of plant biomass in oligotrophic marine environments, that DIN promotes the growth of algae and the PON primarily promotes the growth of the animal.

Our results for ammonium enrichment have recently been confirmed by Hoegh-Guldberg & Smith (1988). They found that addition of ammonium caused an increase in population density of zooxanthellae in *S. pistillata* from the Great Barrier Reef. Interestingly, they found no significant effect of ammonium on zooxanthellae density in *Seriatopora hystrix*.

The ability of corals to take up and assimilate ammonium is well established from previous investigations (see, for example, Burris 1983). Our results reveal that addition of ammonium is accompanied by a significant decrease in the C:N ratio of zooxanthellae, from 7.7 in untreated corals to 4.2–4.7 in N-, N+P- and *Artemia*+N-treated corals. Smith & Muscatine (1986) observed a similar trend in C:N of zooxanthellae in fed and starved polyps of *Aiptasia pulchella* subjected to ammonium enrichment. The significant linear correlation between zooxanthellae N and protein (figure 3) suggests that ammonium is taken up by the zooxanthellae and that uptake is closely coupled to algal protein synthesis. It would appear that algal protein synthesis *in vivo* is N limited. Moreover, there is no statistically significant effect of DIN on animal protein levels (LSD, $p > 0.90$) so that the direct assimilation of ammonium by the animal is interpreted as either negligible or not involved in animal protein synthesis. These results also imply that if organic nitrogen is translocated to the animal from the algae it does not result in a significant change in level of protein.

The effect of adding P with N on zooxanthellae population densities is not significantly different from that of adding N alone. Because there was no effect of P alone on biomass parameters, we can conclude that N and P did not act synergistically during the treatment period (14 days). As addition of P would be expected to promote more nucleic acid synthesis, were it limiting, a tentative conclusion from these results is that zooxanthellae protein synthesis is not limited at the transcriptional level *in vivo*. It seems more likely that zooxanthellae protein synthesis is limited at the translational level, presumably by limitation of the rate of supply of amino acids. Alternatively, there is the remote possibility that added

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phosphate may not have been 'available' because phosphate is known to bind to calcium (cf. Entsch *et al.* 1983), even though the exposed portion of the base of the coral skeleton was small relative to the total skeletal surface.

Feeding on *Artemia* alone or together with P had no significant effect on the host biomass or the population density of zooxanthellae. Although we observed ingestion of *Artemia* during the 1 h feeding periods, we have no way of ascertaining either the number of *Artemia* that were ingested and assimilated or the timecourse and extent of release of catabolic ammonium. It could be that too few *Artemia* were ingested to affect either the nutrient demand of the algae or the growth of the animal.

It might be expected that colonies supplied with *Artemia* + N and *Artemia* + N + P would manifest an increase in population density of the zooxanthellae comparable to colonies supplied with N or N + P alone. Whereas both *Artemia* + N- and *Artemia* + N + P-treated colonies had greater average population densities than untreated colonies or colonies fed *Artemia* with or without P, the differences after exposure for two weeks are not significant at the 90% level. There are several explanations for this result. The uptake of both ammonium and phosphate by whole corals follows a 'depletion–diffusion' model (D'Elia *et al.* 1983), in which inorganic nutrient diffuses through the nutrient-depleted animal tissue to the algae where it is actively transported and assimilated. When uptake kinetics of inorganic nutrients for intact corals are corrected for diffusion, uptake follows Michaelis–Menten kinetics. The half-saturation constants for ammonium assimilation range from 0.3 to 1.05 μM in a variety of corals (Muscatine & D'Elia 1978). Uptake is dependent on the concentration, not the flux, of ammonium. Thus colonies fed *Artemia* would be expected to have locally elevated ammonium concentrations, above the value of K_s for zooxanthellae, which would effectively retard the influx of DIN. Alternatively, assimilation of *Artemia* + N may have resulted in coral growth and increase in numbers of zooxanthellae, so that the net effect was an insignificant change in zooxanthellae density. Although this interpretation is not supported by the data on animal protein in table 1, the effect of *Artemia* + N should be investigated further by using larger sample sizes of corals in which ingestion and assimilation of *Artemia* is clearly established.

The addition of ammonium did not appear to cause a significant increase in MI of the zooxanthellae at the end of incubation for 14 days. However, MI data were obtained only at the end of the experiment, and we could have missed an episode of high growth rate such as that reported by Cook *et al.* (1988) for zooxanthellae in *Aiptasia pallida*. Also, given the small values of MI for coral zooxanthellae compared with those found in *A. pallida*, small changes in MI would be sufficient to cause large changes in specific growth rate, but they would not be detected because the signal:noise ratio is so low in these measurements (compare MI data with calculated u in table 2).

We calculated the average specific growth rates of zooxanthellae from each treatment based on the observed mitotic indices, with the assumption that cell division is asynchronous and that the average duration of division is 0.46 days (= 11 h) (Wilkerson *et al.* 1983). Specific growth rates ranged from 0.080 per day in N + P-treated corals to 0.026 per day in corals exposed to *Artemia* + N (table 2).

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We also calculated the apparent growth rate of zooxanthellae in treated corals from the data on initial (control) and final numbers of cells. Over the exposure for 14 days to DIN zooxanthellae populations approximately doubled, and using the average data for N- and N+P-treated corals, we calculate a net growth rate of 0.045 per day, which is not significantly different from the average specific growth rate of 0.058 ± 0.018 per day estimated from mitotic indices ($p < 0.10$). Rahav *et al.* (1989) examined the recycling of internal nitrogen between the zooxanthellae and animal host. The results of that study suggest that turnover time of nitrogen in *S. pistillata* zooxanthellae is approximately four days. That is, the algae assimilate and excrete the equivalent of their nitrogen-specific biomass at a rate of 0.25 per day. As the specific growth rate of the algae *in vivo* is 10- to 30-fold lower than the turnover time of nitrogen, it would appear that the flux of nitrogen is sufficient to sustain cells near u_{max}, assuming that losses of nitrogen from the coral are small. Thus inorganic nitrogen does not appear to alter the rate of growth of the algal cells, but does allow more cells eventually to accumulate within the host.

In another companion study of the effects of N, N+P and *Artemia* on the photosynthetic characteristics of *S. pistillata*, we found that the flux per unit area of oxygen at saturating irradiance did not increase in proportion to the increase in density of zooxanthellae. Indeed, the maximum rate of photosynthesis per zooxanthella decreased (L. R. McCloskey *et al.*, personal communication). These results suggest that the light-saturated photosynthetic rates are not limited in *S. pistillata* by the concentration of zooxanthellae, but are more likely to be limited by the availability of dissolved inorganic carbon (Dennison & Barnes 1988; Muscatine *et al.* 1989; ter Kuile *et al.*, personal communication).

Previously, it has been shown that growth of zooxanthellae *in vivo* is not coupled to photosynthetic rate; only 1–5% of the carbon fixed by the cells is allocated to new cell material (Muscatine *et al.* 1984; Falkowski *et al.* 1984). Most of the remaining carbon fixed is translocated to the animal host, where it is respired or released as dissolved organic carbon. This acquisition of carbon with little apparent gain in biomass has led to the so-called 'junk-food' hypothesis (Falkowski *et al.* 1984), which proposes that the translocated carbon is rich in calories but deficient in materials that are required to synthesize new cells. Upon addition of inorganic nitrogen, however, more carbon is allocated to amino acids and other nitrogen-rich organic molecules (e.g. glutamine) that appear to be largely retained by the algae (Muscatine 1980*b*), and which become incorporated into new cells. Less carbon is then available for translocation to the host tissue. Therefore not only is the population density of zooxanthellae limited by the availability of inorganic nitrogen, but also the amount of fixed carbon translocated from the zooxanthellae to the animal is inversely related to the zooxanthellae growth rate (see Hoegh-Guldberg & Hinde 1986; Smith 1986). L. R. McCloskey *et al.* (personal communication) calculated that the addition of DIN reduced the level of translocated carbon from 97% of the total daily fixed to approximately 80%.

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Conclusions

For decades coral-reef biologists have noted that the population density of zooxanthellae within a coral species is remarkably constant, in spite of exposure to apparently large gradients in light and temperature. Our experimental results suggest that in *S. pistillata* the population density of zooxanthellae can be affected by the availability of dissolved inorganic nitrogen. Unlike free-living unicellular algae (see Falkowski *et al.* 1984), photosynthesis by zooxanthellae is very poorly coupled to growth, and growth irradiance plays a minor role in determining the population densities. It would appear that the organic nitrogen acquired by the animal host by feeding has little or no effect on the accumulation of algae unless meted to them as an inorganic end-product of nitrogen catabolism. In this sense, the heterotrophic acquisition of organic nitrogen may be essential for the maintenance of the animal, but not necessarily the algal, component in this symbiotic association.

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